# ECON 895 Lab Exam, What to Study

Spatial Techniques in Empirical Economics

Monday 5 October 2026  7:20 to 8:30 PM  Research Hall 201

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**When.** The exam takes the first seventy minutes of class. We break afterwards and the Week 6 lecture starts at 8:45, so the evening runs its usual length. Arrive on time. Nobody gets extra minutes at the end for a late start.

**The rules.** Closed book and closed model. No notes, no internet, no generative AI of any kind. R's own help is permitted, so `?st_buffer` is fine and a browser is not. This is the one assessment in the course where the tools are off, and using them is an honor code matter rather than a grading one.

**What to bring.** Your laptop, with the course repository pulled and up to date before you walk in. Open `ECON895.Rproj` first so your paths resolve. Everything the exam reads is already in `data/`, and you will not need to download anything.

Do this on Sunday rather than Monday evening. Run `git pull`, then run `code/00_setup_test.R` and confirm every check passes. A laptop that cannot read the data at 7:20 on Monday is the one failure I cannot repair in the room.

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## The shape of the paper

Five tasks, ten points. Three are code and two are written. The code tasks carry two, two, and three points. The written tasks carry two and one.

You will get a starter script with the tasks laid out and space under each one. Every task is printed in that script, including the two written ones, so the file you submit is the whole exam and the whole submission. You write your code under each task and your written answers as comments in the same place. You submit that one `.R` file at 8:30, and it should run in a fresh session without errors.

The handout also carries a list of functions you may need for the code tasks. It is longer than the exam requires, it is grouped by purpose rather than by task, and some entries on it are wrong for what you are being asked to do. Nothing on this exam turns on whether you remembered a name. Choosing between two that sound alike, and saying why, is the part that earns marks.

Someone who has worked the weekly scripts by hand finishes the code in about forty minutes. The rest of the time is for the written parts and for checking what you wrote.

## What it draws on

Weeks 1 to 5. The code tasks lean on the vector workflow from the first weeks. The written tasks lean on the raster material from Week 4 and the making-data material from Week 5, which we cover on 28 September. You will not be asked to write raster code under time pressure, and you will not be asked to click through QGIS.

Here is what you should be able to do without looking anything up.

### From the vector weeks

- Read a table of coordinates and turn it into an `sf` object, naming the coordinate reference system explicitly rather than hoping.

- Know which of your two columns is longitude and which is latitude, and know how to check before you measure anything with them.
- Measure a distance between features and report it in the units you intended.
- Choose a projection suited to the operation, and say why that one rather than the one beside it.
- Join two tables and account for every row on both sides, going in and coming out.
- Build a buffer, and know what units the distance you handed it is in before you believe the result.
- Name the spatial predicate you used and say why it is the right one.
- Count how many features fall inside a polygon, and know what that count does and does not include.

**From the raster week**

- Read a raster header and say what it tells you, and what it leaves out.
- Say what a degree grid does to cell area as you move away from the equator, and what that does to an average.
- Say which resampling method a categorical layer needs, and what the wrong one produces.
- Say what `exact = TRUE` changes in a zonal statistic and when the difference is large enough to matter.

**From the making-data week, on 28 September**

- Say what a georeferencing residual reports, and what it stays silent about.

**The recurring habits**

You have met most of these at the top of a deck more than once. They are what the grading rewards.

- Count rows on both sides of every join, and then look at what you kept.
- `st_set_crs()` relabels and `st_transform()` moves. Know which one you meant.
- Equal-area before `st_area()`. Units are a free check that catches only some of the errors.
- Plot it. Your eye catches a coordinate in the wrong ocean faster than any summary will.
- A number extracted from a surface is a measurement you designed.

## How it is graded

Partial credit is generous for correct reasoning with a wrong number, and sparing for a right number with no visible check. That asymmetry is the whole course, and the exam is where it becomes a grade.

Two things follow from it, and they are worth more to you than any amount of syntax revision.

1. **Show the check.** If you count something, report the count before and after. If you measure something, say what units you are in. A check you ran and wrote down earns marks on its own.
2. **Write down what you meant.** If the syntax will not come, put a comment where the code should be, saying what you intended to do and why that would have been the right thing. Reasoning earns marks. A blank earns none.

## How to prepare

1. Work `code/week01.R` through `code/week04.R` again, by hand, in a fresh session, with the slides closed. Typing them is the point. Reading them is not.

2. Do the blocks marked PREDICT first, and commit to an answer before you run each one. Those blocks are the traps, and the traps are where the marks are.

3. When a block surprises you, stop and write one sentence in your own words about why. That sentence is most of a written answer.

4. Week 5's script arrives on 28 September. Work it the same way that week rather than the night before.

5. Try one operation on data you have not used before. If you can build points from a new table, put them in a sensible projection, and count how many land inside a polygon, you are ready.

## What is not on it

No regression is estimated anywhere on this exam. There is no instrumental variable, no regression discontinuity, and no difference in differences to fit. Those designs are what the methods are for, and they are assessed in the coding exercises and the paper, where you have your tools and your time.

You will not be asked to recall a function name or its arguments from memory. The handout lists the candidates and R's help is open. You will not be asked to reproduce a number from a lecture. You will not be asked to operate QGIS.

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The exam is short, it is early in the evening, and it is worth ten percent. It exists to make one argument concrete. What you write yourself in September is what you will be able to check a machine on in October, and the weeks after this one lean on these habits harder than these weeks did. Come and see me in office hours before 5 October if any of the above is shaky.